

INTAKE-ARBITER -- FREE-COOLING DECISION REPORT

An agent that decides, hour by hour, whether outside air can cool a data centre.

THE SITE

Location Amazon IAD103, VA -- station KHEF, America/New_York
Committed pair OSM 996899487 -> 1227612684
Amazon IAD101 / Amazon IAD103
Facade gap 150.4 m between the two halls
Weather record 42,554 real hourly records from KHEF
Plume physics 576 steady-state solves on this site's own footprints; worst intake rise 0.3701 C at 270 deg

FortyGuard data: weather and geometry are this site's own; the four measured FortyGuard level offsets and the N-26 coverage record are Ashburn's, because only Ashburn has forecast/outcome day pairs. Quote the hours for this site; quote the coverage for Ashburn.

THIS REPORT IS A SNAPSHOT, NOT THE LIVE PAGE

The interface recomputes the decision for whatever configuration a reader selects. This file was generated at build time for the ONE configuration named below, chosen because it is the configuration in which this site's agent does the most while still changing mode at least once. If the numbers on screen differ from the numbers here, the configuration differs -- compare the two lists before concluding anything else.

Day 2023-04-11 (crossing)
Plant limit 18.0 C
Notice required 3 h
Forecast skill 0.50 relative to persistence
Level anchor sensor
Condenser bank longest facade
Switch budget 1 mode changes per day
Minimum dwell 3 h
Max dew point 15.0 C (Green Grid WP#46 p.6)

WHAT THE AGENT DID ON THIS DAY

The agent ran free cooling for 3 of 24 hours with 1 mode change. The reactive incumbent took 12 hours -- 9 more -- but declared 1 unsafe hour against the agent's 0. The incumbent had to break its own switch budget 1 time to stay safe; the agent never did. 11 hours were safe but still ran chillers, because the schedule could not afford them.

Agent 3 of 24 hours free cooling, 1 mode change(s), 0 unsafe hour(s)
Incumbent 12 of 24 hours free cooling, 2 mode change(s), 1 unsafe hour(s)
the incumbent broke its own switch budget 1 time(s) to stay safe
11 hour(s) passed every gate and still ran chillers, because the schedule could not afford them

EVERY HOUR, AND THE REASON FOR IT

hh	mode	safe	binding	bound	limit	actual	margin
00	mechanical	yes	switch budget	5.006	18.0	5.002	0.874
01	mechanical	yes	switch budget	4.743	18.0	3.894	1.023
02	mechanical	yes	switch budget	3.913	18.0	2.224	0.919
03	mechanical	yes	switch budget	4.193	18.0	2.784	0.878
04	mechanical	yes	switch budget	2.803	18.0	1.114	0.824
05	mechanical	yes	switch budget	1.438	18.0	0.004	0.790
06	mechanical	yes	switch budget	1.938	18.0	-0.606	1.095
07	mechanical	yes	switch budget	4.468	18.0	3.894	1.402
08	mechanical	yes	switch budget	8.358	18.0	8.894	2.131
09	mechanical	yes	switch budget	11.371	18.0	12.780	2.563

10	mechanical	yes	switch budget	15.858	18.0	16.674	2.687
11	mechanical	no	dry-bulb	18.896	18.0	20.000	2.054
12	mechanical	no	dry-bulb	21.138	18.0	22.784	1.517
13	mechanical	no	dry-bulb	22.800	18.0	23.890	1.183
14	mechanical	no	dry-bulb	24.185	18.0	24.440	1.069
15	mechanical	no	dry-bulb	25.574	18.0	25.922	0.974
16	mechanical	no	refusal	25.538	18.0	25.040	1.041
17	mechanical	no	refusal	24.735	18.0	24.445	1.187
18	mechanical	no	refusal	24.232	18.0	22.781	1.484
19	mechanical	no	dry-bulb	21.408	18.0	18.334	1.633
20	mechanical	no	dry-bulb	19.748	18.0	15.564	1.745
21	FREE	yes	-	17.248	18.0	12.224	1.595
22	FREE	yes	-	14.743	18.0	11.114	1.476
23	FREE	yes	-	11.968	18.0	8.334	1.129

bound = the agent's 90 %-nominal upper bound on intake air, forecast plus margin

actual = what the intake temperature turned out to be, from the station record

margin = group-conditional forecast error for that hour of day, plus plume spread

THE REASONING, HOUR BY HOUR

00:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 5.006 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

01:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 4.743 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

02:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 3.913 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

03:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 4.193 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

04:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 2.803 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

05:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 1.438 C

against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

06:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 1.938 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

07:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 4.468 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

08:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 8.358 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

09:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 11.371 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

10:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 15.858 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

11:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 18.896 C against a 18.0 C limit, so it fails by 0.896 C. It would take a limit 0.896 C higher, or a bound 0.896 C tighter, to change this hour.

12:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 21.138 C against a 18.0 C limit, so it fails by 3.138 C. It would take a limit 3.138 C higher, or a bound 3.138 C tighter, to change this hour.

13:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 22.800 C against a 18.0 C limit, so it fails by 4.800 C. It would take a limit 4.800 C higher, or a bound 4.800 C tighter, to change this hour.

14:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 24.185 C against a 18.0 C limit, so it fails by 6.185 C. It would take a limit 6.185 C higher, or a bound 6.185 C tighter, to change this hour.

15:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 25.574 C against a 18.0 C limit, so it fails by 7.574 C. It would take a limit 7.574 C higher, or a bound 7.574 C tighter, to change this hour.

16:00 MECHANICAL [refusal]

Mechanical, and the reason is a REFUSAL rather than a temperature. At this wind bearing a building sits between the condensers and the air intake. The dispersion model has no representation of a building standing in the flow, so any number it produced here would be meaningless. The agent declines to certify the hour instead of returning a figure it cannot stand behind.

17:00 MECHANICAL [refusal]

Mechanical, and the reason is a REFUSAL rather than a temperature. At this wind bearing a building sits between the condensers and the air intake. The dispersion model has no representation of a building standing in the flow, so any number it produced here would be meaningless. The agent declines to certify the hour instead of returning a figure it cannot stand behind.

18:00 MECHANICAL [refusal]

Mechanical, and the reason is a REFUSAL rather than a temperature. At this wind bearing a building sits between the condensers and the air intake. The dispersion model has no representation of a building standing in the flow, so any number it produced here would be meaningless. The agent declines to certify the hour instead of returning a figure it cannot stand behind.

19:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 21.408 C against a 18.0 C limit, so it fails by 3.408 C. It would take a limit 3.408 C higher, or a bound 3.408 C tighter, to change this hour.

20:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 19.748 C against a 18.0 C limit, so it fails by 1.748 C. It would take a limit 1.748 C higher, or a bound 1.748 C tighter, to change this hour.

21:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 17.248 C, which is 0.752 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.595 C, and the margin is measured, not chosen: 1.575 C of group-conditional forecast error for this hour of day, plus 0.0197 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

22:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 14.743 C, which is 3.257 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.476 C, and the margin is measured, not chosen: 1.456 C of group-conditional forecast error for this hour of day, plus 0.0197 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

23:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 11.968 C, which is 6.032 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.129 C, and the margin is measured, not chosen: 1.109 C of group-conditional forecast error for this hour of day, plus 0.0197 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

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